Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_af66175ec51342aba4bd1122cf5fea4a
Date	2026-11-01 18:09:39
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
crack	critical	95%	9.0 mm long

2. Root cause

Cause: Material Hardness Variation

Reason: The detected 9.0 mm crack at the bend radius is directly linked to material hardness variations in SPCC steel, which increases susceptibility to cracking during forming operations as documented in [02cf0928-6739-4f7e-ae1b-f6740283f00f].

Effect: Cracking at bend radius

Confidence: 89%

Supporting evidence
- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba

3. Solution

Corrective actions (fix this occurrence)

1. Verify material certificates against specification for yield strength, tensile strength, and hardness
2. Conduct incoming material hardness testing

Preventive actions (stop it recurring)

1. Implement stricter incoming material quality inspection protocols
2. Audit material supplier quality control processes

Precautions

- Ensure batch traceability for all SPCC steel lots

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Insufficient Evidence for Alternative Causes	86%	The provided evidence [02cf0928-6739-4f7e-ae1b-f6740283f00f] focuses exclusively on material inconsistency. No other evidence regarding tooling, lubrication, or machine parameters was provided to support additional root causes.
3	Work Hardening Inconsistency	83%	The evidence [02cf0928-6739-4f7e-ae1b-f6740283f00f] identifies work hardening inconsistency as a contributor to surface defects and cracking in SPCC steel, which aligns with the critical crack detected.

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.